Lecture V: Randomness

Programming: Everyday Decision-Making Algorithms

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Today in one sentence

When a problem is too big to solve exactly, a bit of randomness often gets you a good answer fast.

Why it matters

Will your project finish on time? Which route through 15 cities is shortest? Questions like these are too complex to calculate exactly. Companies use randomness every day to estimate risks, plan routes and allocate scarce places fairly.

Roadmap

1. Randomness everywhere: why chance is useful
2. Simulate instead of calculate: Monte Carlo
3. Let chance help you decide: smarter searches, fairer choices

Randomness everywhere: why chance is useful

What comes to your mind?

Question: **What comes to your mind when you think of randomness?**

Casino

Lottery

Shuffling cards

Algorithms

Cryptography

Genetic mutations

The opposite of randomness is predictability

Question: **What's the opposite of randomness?**

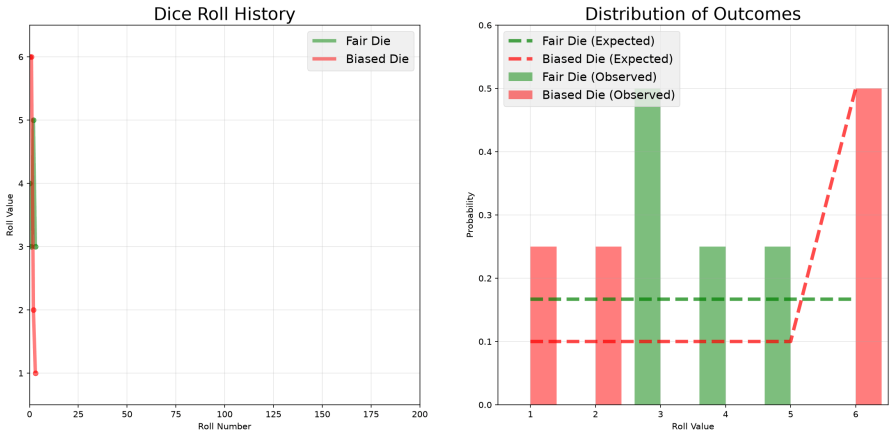
- **Determinism** - same input, same output
- **Predictability** - knowing the outcome in advance
- **Consistency** - reliable patterns
- Boredom?

Is the die fair? Roll it many times

Question: **How would you test if a pair of dice is fair?**

- Send the dice **to a lab** to check weight and balance
- Roll the dice **many times**
- Check if the outcomes are **uniformly distributed**
- Compare **observed** frequencies to **expected** frequencies

A fair and a loaded die after 200 rolls



Randomness is a tool, not just a gamble

- **Testing:** random inputs find bugs
- **Optimization:** random steps escape dead ends
- **Simulation:** model systems with uncertainty
- **Security:** stay unpredictable for attackers
- **Fairness:** remove bias from selection

Where do you meet randomness every day?

Question: **Where do you encounter randomness in daily life?**

Social life: chance brings people together

- **Dating apps:** Randomized matching within preferences
- **Random encounters** that lead to friendships
- **Career opportunities** from unexpected connections
- **Breaking ties** through coin tosses
- **Netflix recommendations:** Controlled randomness for discovery

Entertainment: controlled randomness keeps us playing

- **Pokémon:** “Random” encounters weighted by rarity
- **Loot systems:** Rare items have controlled drop rates
- **Chess AI:** Introduces randomness to feel more human-like
- **Spotify shuffle:** Deliberately less random to feel natural
- **TikTok:** Controlled randomization for content discovery

Security: unpredictability protects us

- **Cryptocurrency mining:** Solving cryptographic puzzles through guessing
- **Password generators:** Balance randomness and memorability

- `correct-horse-battery-staple` is more secure than `Tr0ub4dor&3`
- **Modern encryption** relies on random number generation
- **Two-factor authentication** uses random codes

Research: random samples give unbiased answers

- **Weather forecasting:** randomness models uncertainty
- **Stock algorithms:** randomness avoids predictable patterns
- **Self-driving cars:** random elements feel natural
- **Random sampling:** unbiased research results
- **A/B testing:** random assignment to groups

Why computer scientists love randomness

- **Efficiency:** often faster than deterministic approaches
- **Simplicity:** complex problems, simple code
- **Robustness:** avoids worst-case scenarios
- **Approximation:** "good enough" solutions in reasonable time

True randomness comes from physics, pseudo-randomness from a seed

Question: **Difference between true and pseudo-randomness?**

True Randomness

- Physical phenomena
- Atmospheric noise
- Radioactive decay
- Quantum effects

Pseudo-randomness

- Deterministic algorithms
- Seed-based generation
- Repeatable sequences
- Good enough for most uses

Randomness everywhere: why chance is useful

What we covered

Randomness is more than gambling: it makes algorithms faster, systems secure and discoveries possible.

Simulate instead of calculate: Monte Carlo

Roadmap

1. Randomness everywhere: why chance is useful

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Some things are too big to count

Question: **How many possible orders exist for a shuffled deck of cards?**

```
import math
print(math.factorial(52))
```

```
80658175170943878571660636856403766975289505440883277824000000000000
```

> ! Important
>
> Computing and evaluating all possible combinations is not feasible!

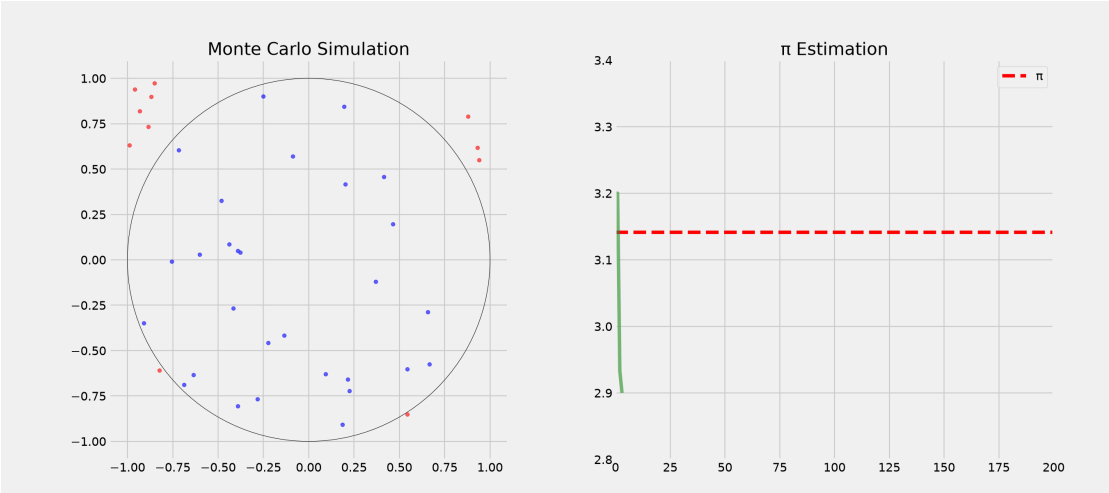
Monte Carlo: estimate by running many random trials

Question: **Anybody ever heard of "Monte Carlo methods"?**

- 1940s nuclear research: chain reactions were too complex to calculate
- Idea: draw random cases many times and count the outcomes
- Named after **Monaco's famous casino**

Question: **How could we estimate π?**

Estimating π by throwing random points

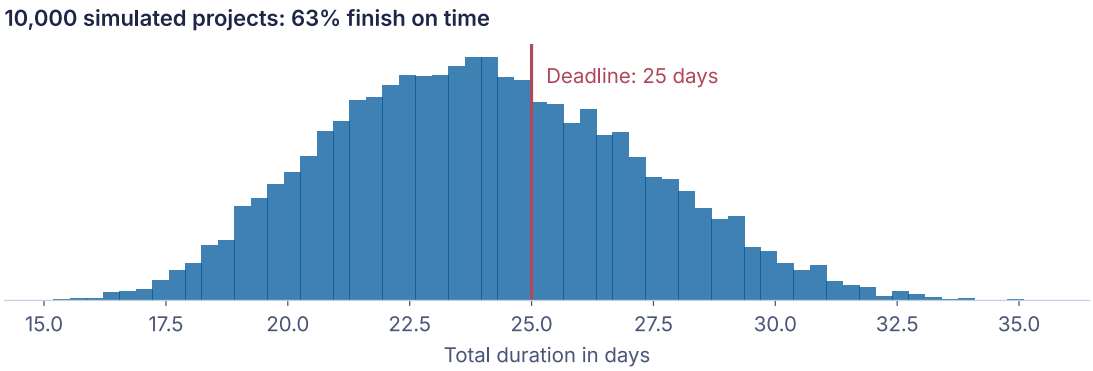


Monte Carlo in business: a range instead of a single forecast

- **Project planning:** how likely do we finish before the deadline?
- **Finance:** how much could a portfolio lose in a bad month?
- **Inventory:** how much stock covers an uncertain demand?
- **Recipe:** model the uncertain inputs, draw random values many times

Most likely 21 days, but only 63 % finish by day 25

Three uncertain tasks, most likely **21 days** in total, deadline **25 days**:



Simulate instead of calculate: Monte Carlo

What we covered

Running many random simulations gives reliable estimates where exact calculation is out of reach.

Let chance help you decide: smarter searches, fairer choices

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Ten cities, one trip: which order is shortest?

Question: **In which order would you visit 10 cities by plane with minimal total distance?**

```
import math
print(math.factorial(10))
```

```
3628800
```

Question: **What could be a strategy?**

Brute force: try every route

- Try every possibility
- Total possible routes: 10! = **3,628,800**
- **Guaranteed** to find best solution

- If each check takes 1ms: **1 hour** to check all routes

Question: **What could be the problem with this approach?**

Twenty cities would take 77 million years

- For 20 cities: 20! = **2.4 quintillion** routes
- About **77 million years** at 1 ms per check
- The effort grows factorially: not feasible for real problems

Greedy: always take the shortest next flight

- Make the **locally best** choice at each step
- Never backtrack or reconsider
- Fast and simple, but can perform **poorly on complex problems**

Hill climbing: improve step by step, until you are stuck

- Improve the solution by small changes, keep the better one
- Like **climbing in fog**: you only see your immediate surroundings
- You get stuck on a **local optimum**, a small hill, and never see the mountain

Question: **How would you escape a local optimum?**

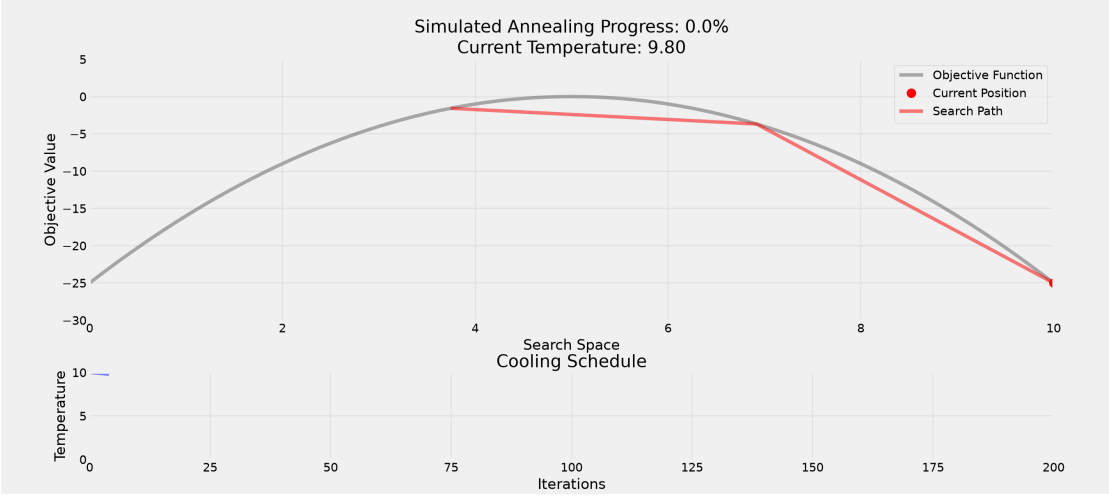
Simulated annealing: sometimes accept a worse step

- Make random changes and **accept improvements**
- Sometimes accept **worse solutions**
- Gradually become **more selective**

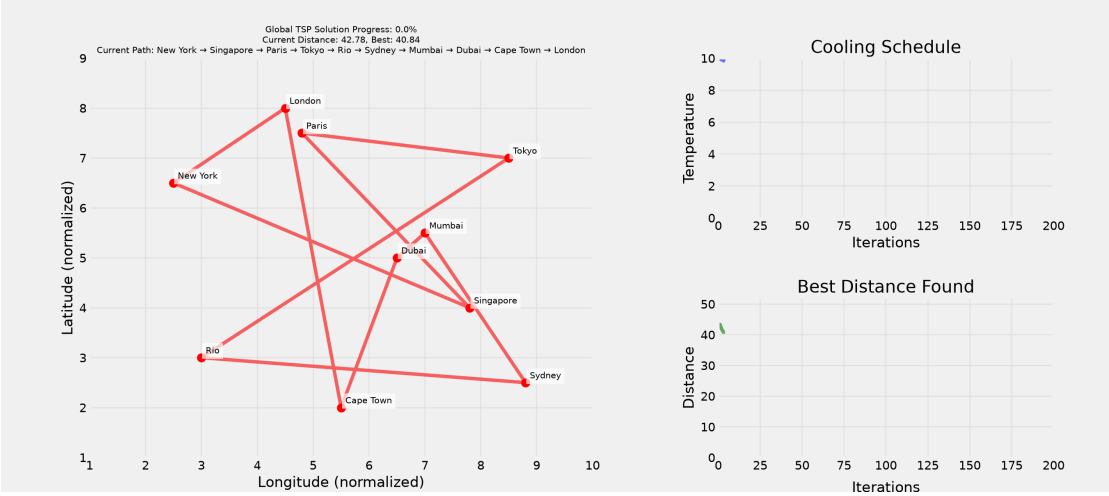
Question: **Why accept worse solutions sometimes?**

- Randomness helps to escape **local optima**
- Balances exploration vs. exploitation

Annealing on a simple curve



Annealing finds a short route through ten cities



Thought experiment: freedom or equality?

What's more important for a society?

Freedom

- Individual choice
- Personal responsibility
- Market-driven

Equality

- Shared resources
- Social safety nets
- Regulated systems

Question: **Any problem with this question?**

Veil of ignorance: design a society without knowing your place in it

You might randomly be:

- Any gender identity and economic status
- Any health condition and intelligence level
- Any cultural background and religious belief

Question: **If you didn't know who you'd be born as, what kind of society would you design?**

Stories mislead, statistics miss nuance

- **Individual stories:** powerful but potentially misleading
- **Statistics:** comprehensive but can miss nuance
- **Hidden diversity:** important subgroups may be overlooked
- **Small policy changes** can have cascading effects

Which measure of success?

- **Mean happiness:** Average well-being
- **Total happiness:** Utilitarian approach
- **Median happiness:** Focus on the middle class
- **Minimum happiness:** Protecting the most vulnerable

Question: **What could be the problem with these measures?**

Idea: ask a random sample

- Randomly select a subset of the population
- Gather **diverse perspectives** from the sample
- Better understand **needs** of population
- **Reduce selection bias** and improve accuracy

Question: **What is a selection bias?**

Selection bias: the sample is not the population

Definition: The sample you analyze is not representative of the population you want to study.

> ⚠ Warning
>
> WWII: engineers wanted to armor returning planes where the bullet holes were. Abraham Wald: armor where there are **no** holes, those planes never returned.

Random choices can be the fairest choices

Question: **How can randomness promote fairness?**

- **Random allocation** of patients in clinical trials
- **Random audits** for tax compliance
- **Random assignment** of cases to judges
- **Random order** of candidates on voting ballots

Quick poll: sure 50 or maybe 120?

Question: **Which would you prefer?**

- **100% chance of winning 50 EUR**
- **50% chance of winning 120 EUR**

> Tip
>
> Answer depends on your **risk aversion**!

When to embrace randomness, when to reduce it

Question: **When should we embrace vs. reduce randomness?**

Embrace When:

- Exploring new solutions
- Avoiding bias
- Breaking deadlocks
- Testing systems

Reduce When:

- Safety-critical systems
- Financial transactions
- Medical procedures
- Legal proceedings

Good enough beats perfect

- **Perfect is the enemy of good:** approximations work, complete analysis is often impossible
- Perfect information is rare
- Quick approximations enable **faster decisions**: balance accuracy against time

Let chance help you decide: smarter searches, fairer choices

What we covered

Random steps help a search escape bad solutions, and random choices can make decisions fairer.

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randomness every day to estimate risks, plan routes and allocate scarce places fairly.

When you can't calculate it, simulate it.

Any questions

so far?

After the break: Randomness

- Tutorial V.I: Programming with AI coding assistants
- Tutorial V.II: Recap of the course topics with randomness
- Monte Carlo simulation of a travel problem in code

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[1]

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Free online
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

> **i** Note
>
> Think Python is free online and a great start. Schrödinger programmiert Python is a playful alternative in German.

More Literature

For more interesting literature, take a look at the literature list of this course.

[1] The main inspiration for this lecture. We have read it and discussed it in depth, always wanting to translate it into a course.